

Tutorial No 5 - FFT (DIT-FFT) and Frequency Response

Aim:- To ~~implement~~^{compute} the DIT-FFT algorithm and evaluate the frequency response of signals.

Theory:-

The N -point DFT of a sequence $x(n)$ converts the time domain N -point sequence $x(n)$ to a frequency domain N -point sequence $X(k)$.

In Decimation ~~In~~ Time (DIT) algorithm, the time domain sequence $x(n)$ is decimated and smaller point DFTs are performed.

The results of smaller point DFTs are combined to get the result of N -point DFT.

In DIT radix-2 FFT, the time domain sequence is decimated into 2-point sequences. For each 2-point sequence, the two point DFT is computed. The results of 2-point DFT are used to compute 4-point DFTs. ~~The~~ A pair of 2-point DFT results are used to compute one 4-point DFT. The results of 4-point DFT are used to compute 8-point DFT's. A pair of 4-point DFT results are used to compute one 8-point DFT. The process is continued until we get N -point DFT.

8-Point DFT using Radix-2 DIT FFT

The input sequence is 8-point sequence. Therefore $N=8=2^3=r^m$. Here $r=2$ and $m=3$

Therefore, the computation of 8-point DFT using radix-2 FFT, involves three phases of computation. The given 8-point sequence is decimated to 2-point sequences. For each 2-point sequence, the 2-point DFT is computed. From the results of 2-point DFT, the 4-point DFT can be computed. From the results of 4-point DFT, the 8-point DFT can be computed.

Let the given sequence be $x(0), x(1), x(2), x(3), x(4), x(5), x(6), x(7)$, which consists of 8 samples. The 8-samples should be decimated into sequences of 2-samples. Before decimation they are arranged in bit reversed order as shown in Table 1.

The $x(n)$ in bit reversed order is decimated into 4 numbers of 2-point sequences as shown below.

Sequence-1 : $\{x(0), x(4)\}$

Sequence-2 : $\{x(2), x(6)\}$

Sequence-3 : $\{x(1), x(5)\}$

Sequence-4 : $\{x(3), x(7)\}$

Table 1.

Normal order		Bit reversed order	
$x(0)$	$x(000)$	$x(0)$	$x(000)$
$x(1)$	$x(001)$	$x(4)$	$x(100)$
$x(2)$	$x(010)$	$x(2)$	$x(010)$
$x(3)$	$x(011)$	$x(6)$	$x(110)$
$x(4)$	$x(100)$	$x(1)$	$x(001)$
$x(5)$	$x(101)$	$x(5)$	$x(101)$
$x(6)$	$x(110)$	$x(3)$	$x(011)$
$x(7)$	$x(111)$	$x(7)$	$x(111)$

Using the decimated sequence as input the 8-point DFT is computed. The Fig. 1. shows the 3 phases of computation of an 8-point DFT

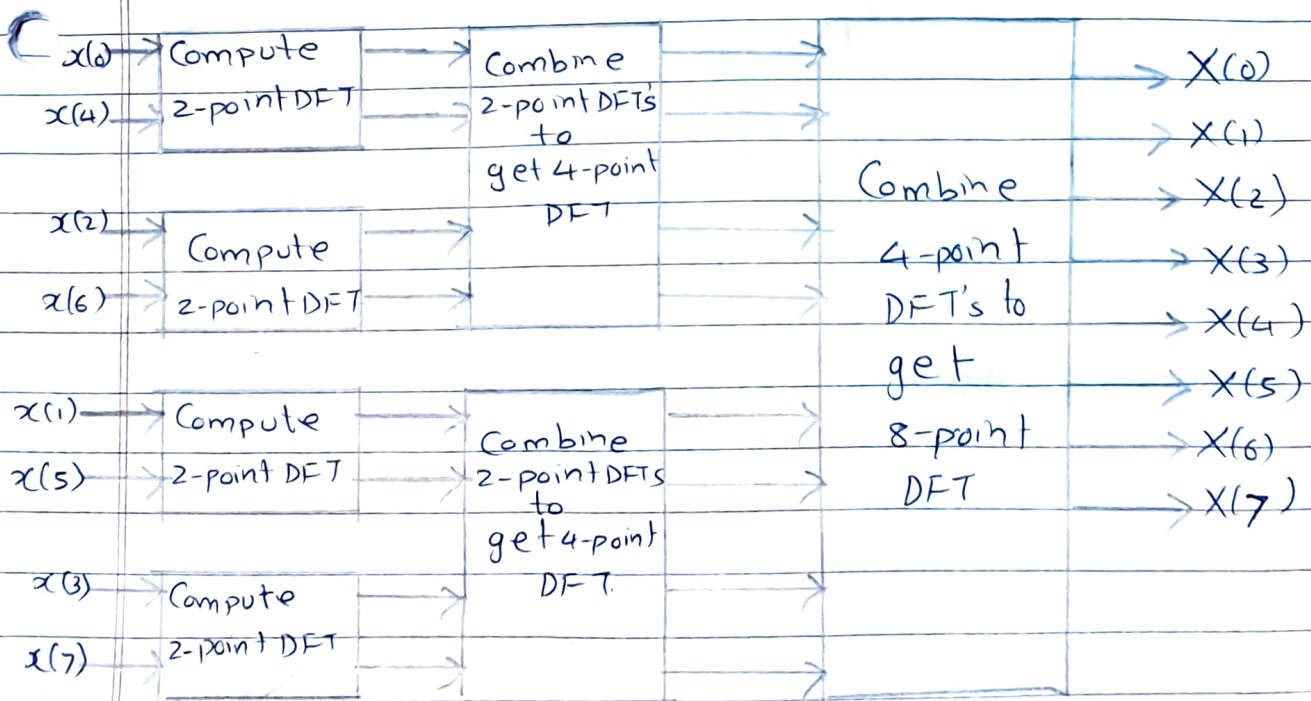


Fig. 1. Three phases of computations in 8-point DFT

The combined Flow Graph (or Butterfly Diagram) of All the Three Phases of Computation

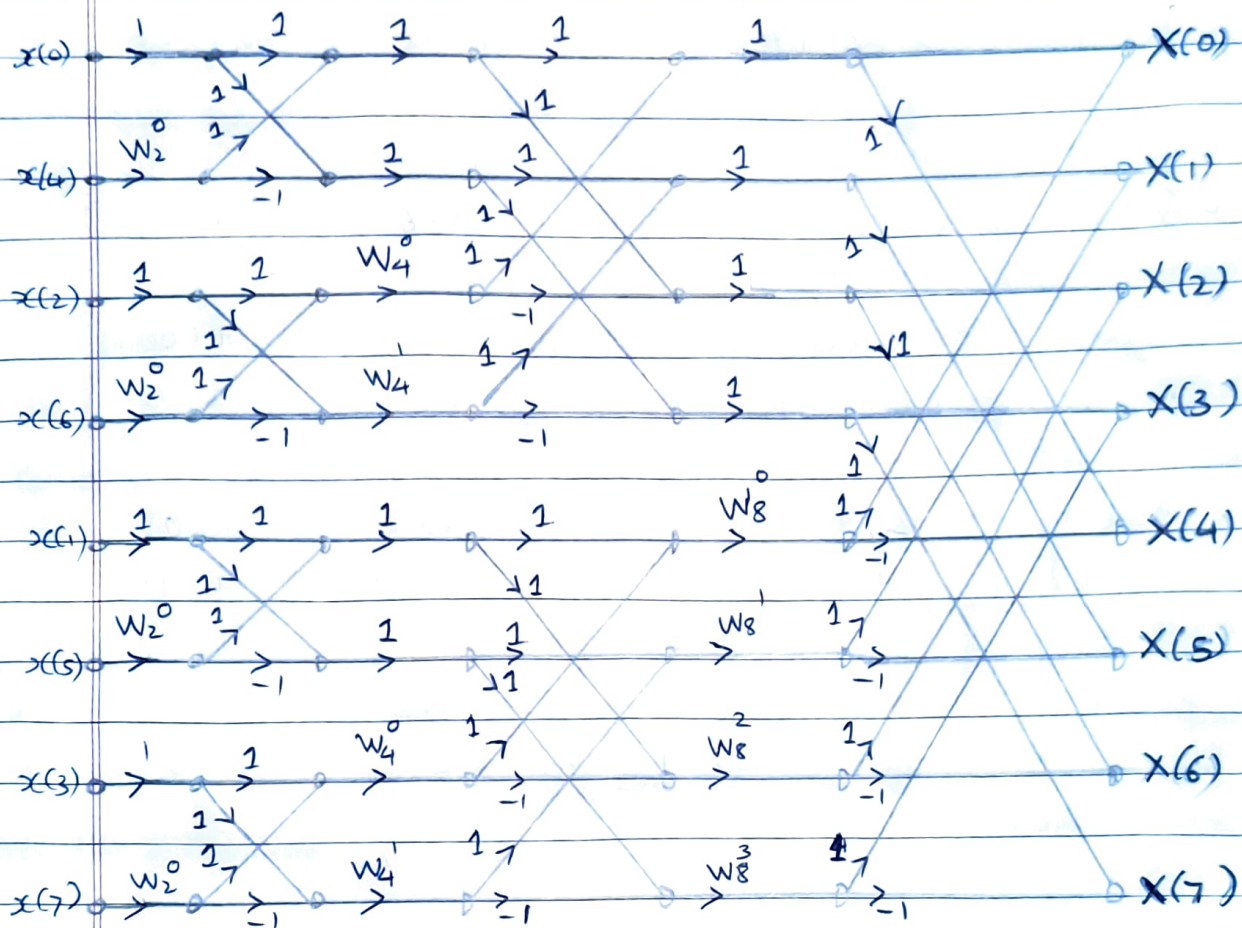


Fig-2. The flow graph (or butterfly diagram) for 8-point DFT via radix-2 DIT FFT.

Problem:-

1) An 8-point sequence is given by $x(n) = \{2, 1, 2, 1, 2, 1, 2\}$.
Compute 8-point DFT of $x(n)$ by

a) radix-2 DIT FFT. Also sketch the magnitude and phase spectrum.

Solⁿ) 8-point DFT by Radix-2 DIT-FFT

The given sequence is first arranged in the bit reversed order

The sequence $x(n)$
in normal order

$$x(0) = 2$$

$$x(1) = 1$$

$$x(2) = 2$$

$$x(3) = 1$$

$$x(4) = 1$$

$$x(5) = 2$$

$$x(6) = 1$$

$$x(7) = 2$$

The sequence $x(n)$
in bit reversed order

$$x(0) = 2$$

$$x(4) = 1$$

$$x(2) = 2$$

$$x(6) = 1$$

$$x(1) = 1$$

$$x(5) = 2$$

$$x(3) = 1$$

$$x(7) = 2$$

The 8-point DFT by radix-2 FFT involves 3 phases of computation with 4-butterfly computation in each phase. The sequence rearranged in the bit reversed order forms the input to the first phase. For other phases of computation the output of the previous phase will be input for the current phase.

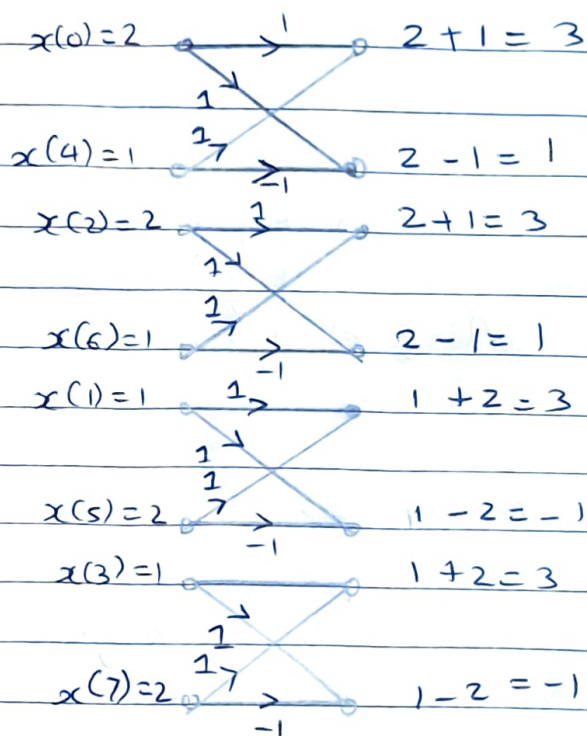


Fig 1 :- Butterfly diagram for 1st phase of radix-2 DIT-FFT

First phase computation:

The input sequence of first phase computation
 $= \{2, 1, 2, 1, 1, 2, 1, 2\}$

The butterfly computations of 1st phase are shown in Fig 1.

The output sequence of first phase of computation
 $= \{3, 1, 3, 1, 3, -1, 3, -1\}$

The phase factor involved in first phase of computation is W_2 . Since, $W_2^0 = 1$, it is not considered for computation.

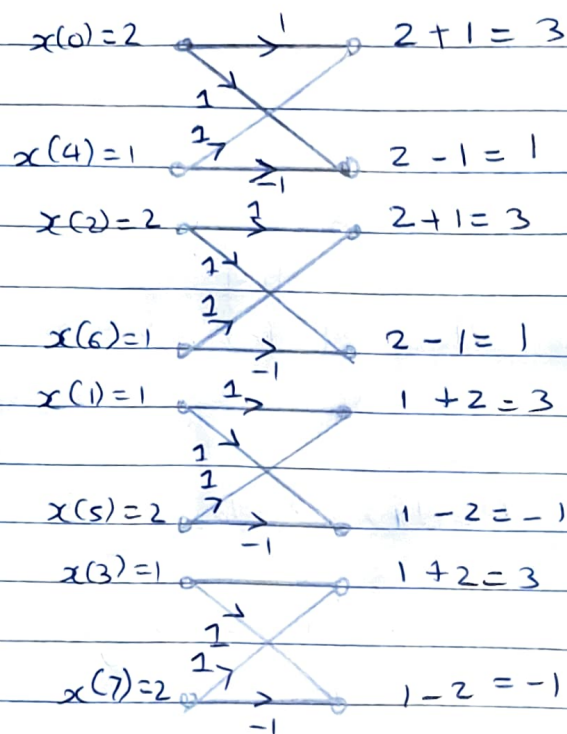


Fig 1:- Butterfly diagram for 1st phase of radix-2 DIT-FFT.

First phase computation:

The input sequence of first phase computation
 $= \{2, 1, 2, 1, 1, 2, 1, 2\}$

The butterfly computations of 1st phase are shown in Fig 1.

The output sequence of first phase of computation
 $= \{3, 1, 3, 1, 3, -1, 3, -1\}$

The phase factor involved in first phase of computation is W_2 . Since, $W_2^0 = 1$, it is not considered for computation.

Second phase computation

The input sequence to second phase computation
 $= \{3, 1, 3, 1, 3, -1, 3, -1\}$

The phase factors involved in second phase are shown in Fig 2.

$$W_4^0 = e^{-j2\pi \times 0/4} = e^0 = 1$$

$$W_4^1 = e^{-j2\pi \times 1/4} = e^{-j\pi/2} \\ = \cos\left(-\frac{\pi}{2}\right) + j \sin\left(-\frac{\pi}{2}\right)$$

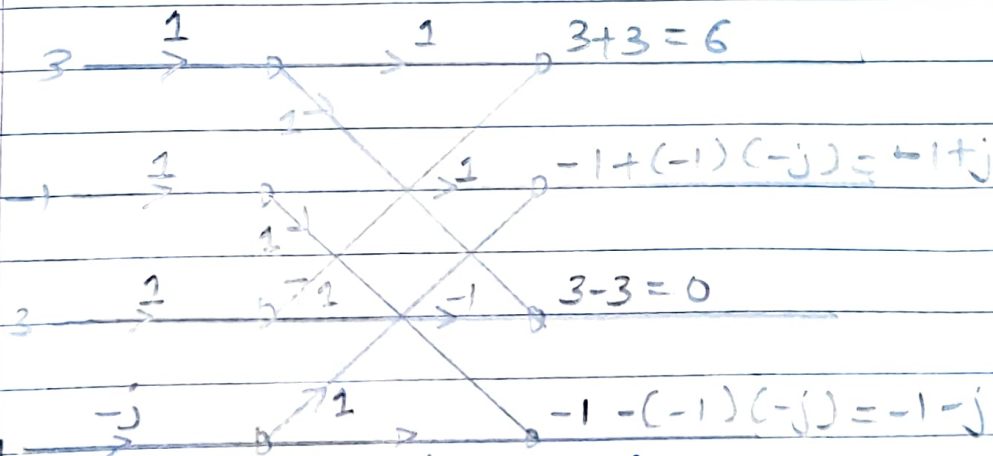
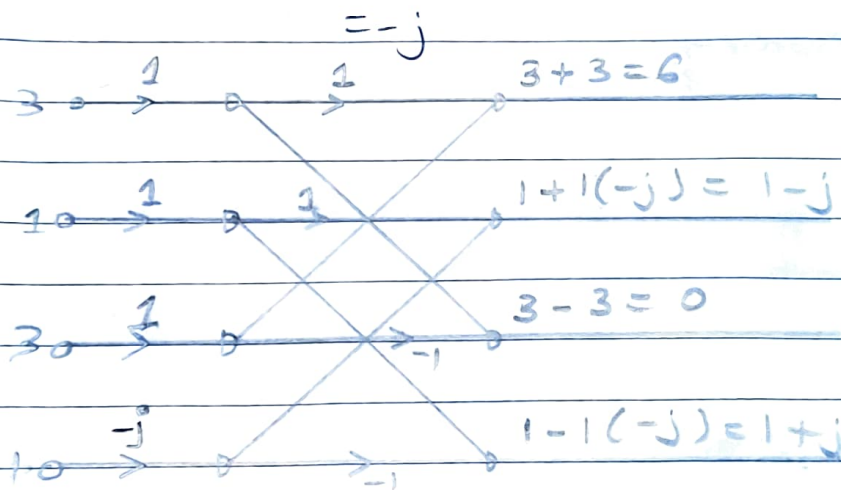


Fig 2: Butterfly diagram for second phase of radix-2 DIT FFT

The output sequence of second phase of computation = $\{6, 1-j, 0, 1+j, 6, -1+j, 0, -1-j\}$

~~Third~~ phase computation

The input sequence to third phase computation = $\{6, 1-j, 6, 1+j, 6, -1+j, 0, -1-j\}$

The phase factors involved in third phase computation are W_8^0, W_8^1, W_8^2 and W_8^3 .

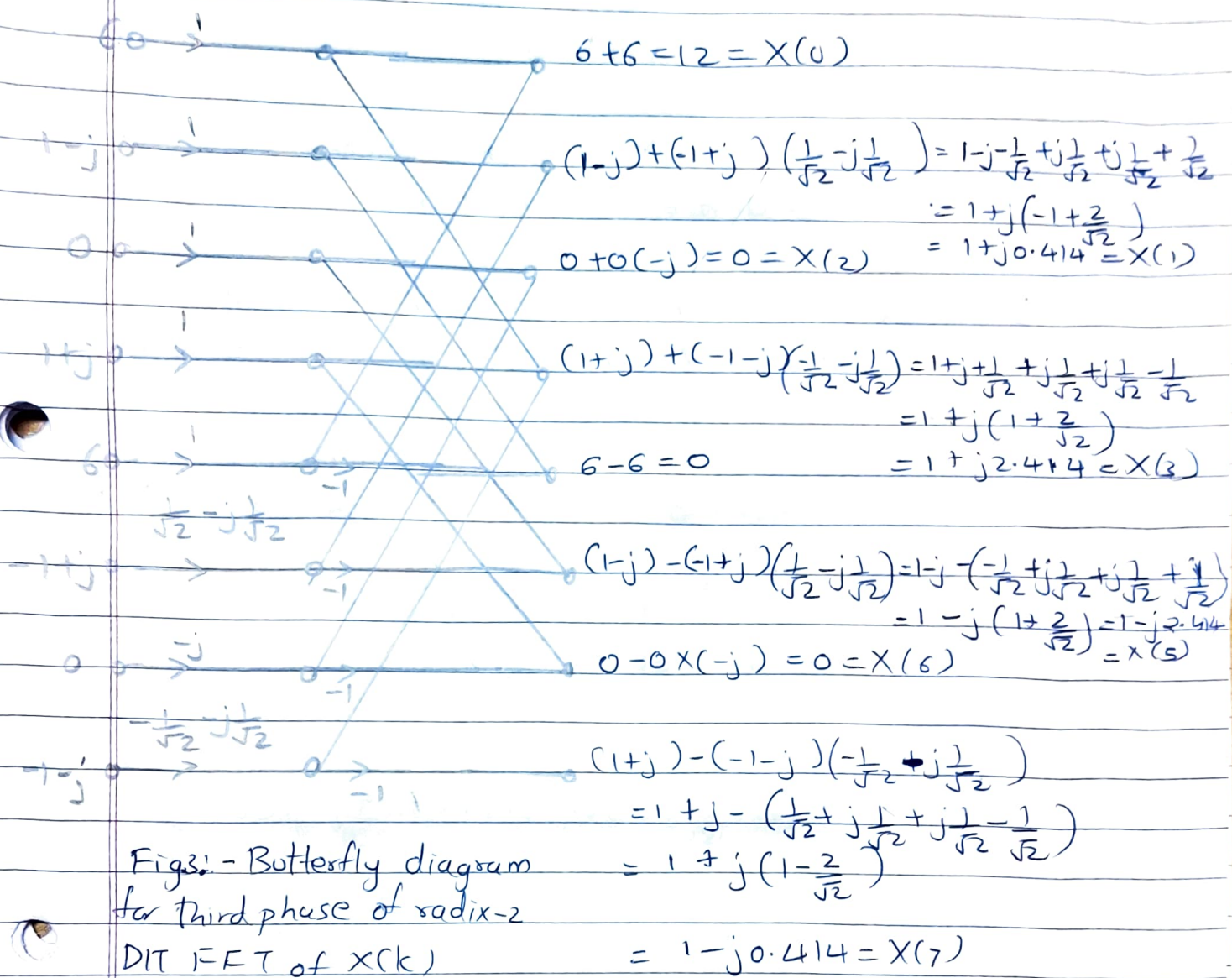
The butterfly computations of third phase are shown in Fig 3.

$$W_8^0 = e^{-j2\pi \times \frac{0}{8}} = 1$$

$$\begin{aligned} W_8^1 &= e^{-j2\pi \times \frac{1}{8}} = e^{-j \times \frac{\pi}{4}} = \cos\left(-\frac{\pi}{4}\right) + j \sin\left(-\frac{\pi}{4}\right) \\ &= \frac{1}{\sqrt{2}} - j \frac{1}{\sqrt{2}} \end{aligned}$$

$$\begin{aligned} W_8^2 &= e^{-j2\pi \times \frac{2}{8}} = e^{-j \times \frac{\pi}{2}} = \cos\left(-\frac{\pi}{2}\right) + j \sin\left(-\frac{\pi}{2}\right) \\ &= -j \end{aligned}$$

$$\begin{aligned} W_8^3 &= e^{-j2\pi \times \frac{3}{8}} = e^{-j \times \frac{3\pi}{4}} = \cos\left(-\frac{3\pi}{4}\right) + j \sin\left(-\frac{3\pi}{4}\right) \\ &= -\frac{1}{\sqrt{2}} - j \frac{1}{\sqrt{2}} \end{aligned}$$



The output sequence of third phase of computation

$$= \{12, 1+j0.414, 0, 1+j2.414, 0, 1-j2.414, 0, 1-j0.414\}$$

The output sequence of third phase of computation is the 8-point DFT of the given sequence in normal order

$$\therefore \text{DFT}\{x(n)\} = X(k) = \{12, 1+j0.414, 0, 1+j2.414, 0, 1-j2.414, 0, 1-j0.414\}$$

Magnitude and Phase spectrum

Each element of the sequence $X(k)$ is a complex number and they are expressed in rectangular coordinates.

If they are converted to polar coordinates then the magnitude and phase of each element can be obtained.

$$X(k) = \{12, 1 + j0.414, 0, 1 + j2.414, 0, 1 - j2.414, 0, 1 - j0.414\}$$

$$= \{12\angle 0^\circ, 1.08\angle 22^\circ, 0\angle 0^\circ, 2.61\angle 67^\circ, 0\angle 0^\circ, 2.61\angle -67^\circ, 0\angle 0^\circ, 1.08\angle -22^\circ\}$$

$$= \left\{ 12\angle 0^\circ, 1.08\angle 22^\circ \times \frac{\pi}{180^\circ}, 0\angle 0^\circ, 2.61\angle 67^\circ \times \frac{\pi}{180^\circ}, 0\angle 0^\circ, 2.61\angle -67^\circ \times \frac{\pi}{180^\circ}, 0\angle 0^\circ, 1.08\angle -22^\circ \times \frac{\pi}{180^\circ} \right\}$$

$$= \{12\angle 0, 1.08\angle 0.12\pi, 0\angle 0, 2.61\angle 0.37\pi, 0\angle 0, 2.61\angle -0.37\pi, 0\angle 0, 1.08\angle -0.12\pi\}$$

$$|X(k)| = \{12, 1.08, 0, 2.61, 0, 2.61, 0, 1.08\}$$

$$\angle X(k) = \{0, 0.12\pi, 0, 0.37\pi, 0, -0.37\pi, 0, -0.12\pi\}$$

The magnitude spectrum is the plot of the magnitude of each sample of $X(k)$ as a function of k as shown in Fig 4.

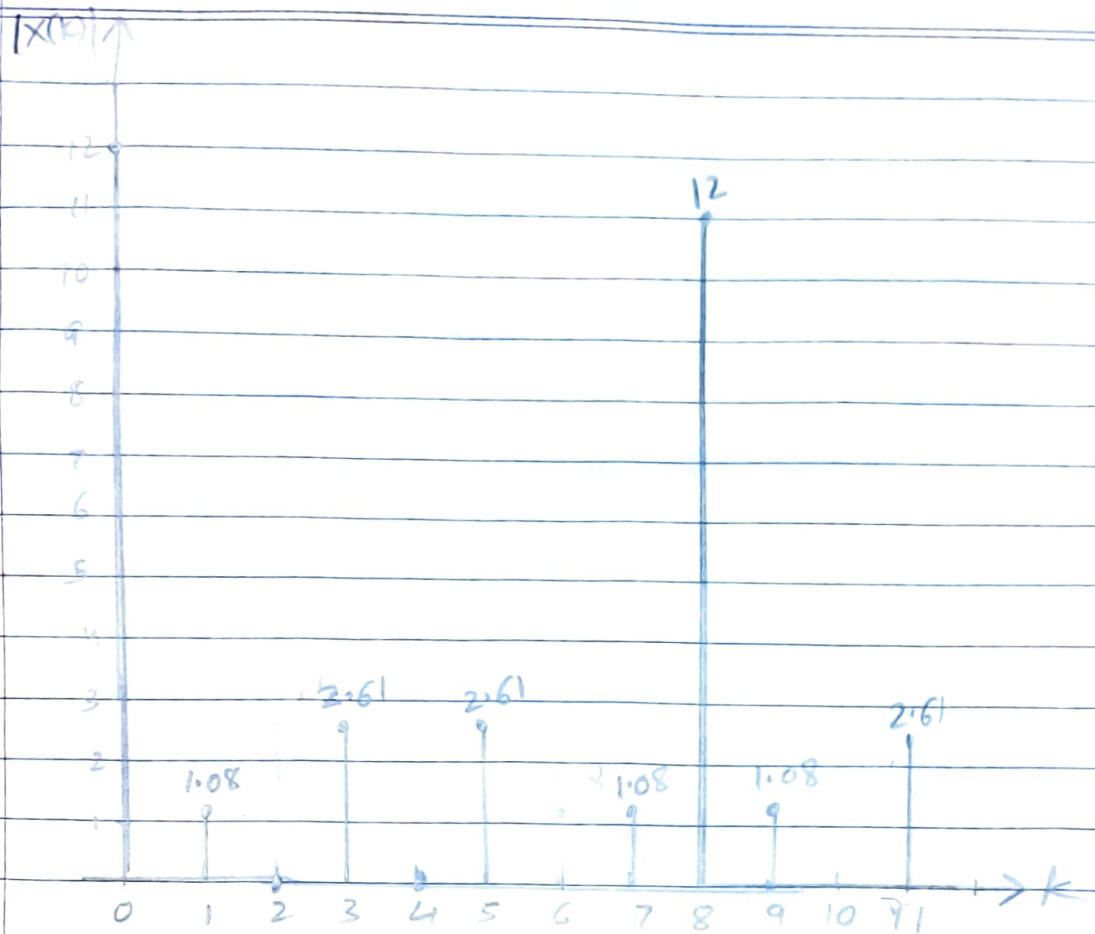


Fig 4: Magnitude spectrum

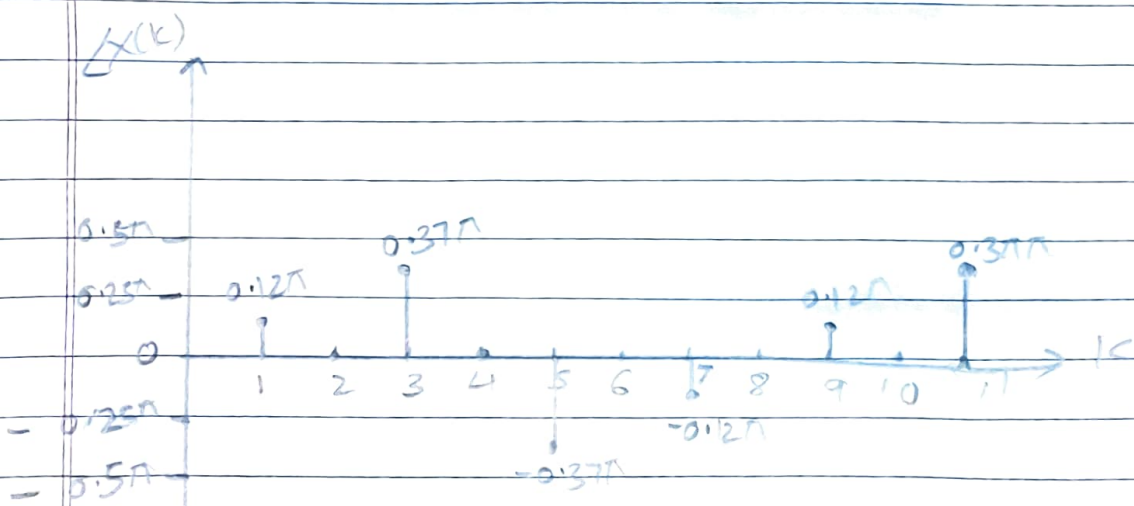


Fig 5:- Phase Spectrum